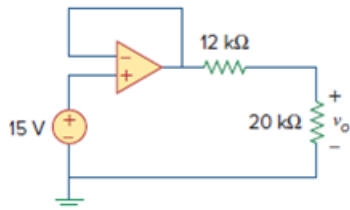


Problem

Calculate v_o in the op amp circuit of Fig. 5.63.

Figure. 5.63:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 5.63 in the textbook.

The non-inverting terminal is directly connected to 15 V voltage source.

The voltage at non-inverting terminal of the op-amp is,

$$v_+ = 15 \text{ V}$$

According to virtual ground concept, the voltages at inverting and non-inverting terminals of the op-amp are same for ideal op-amp.

$$\begin{aligned} v_- &= v_+ \\ &= 15 \text{ V} \end{aligned}$$

The op-amp is voltage follower.

Thus, for a voltage follower

$$\begin{aligned} v_{o(op)} &= v_- \\ &= 15 \text{ V} \end{aligned}$$

Step 2 of 2

Calculate the output voltage using voltage division rule.

$$\begin{aligned} v_o &= \left(\frac{20 \text{ k}\Omega}{12 \text{ k}\Omega + 20 \text{ k}\Omega} \right) (15 \text{ V}) \\ &= \left(\frac{20 \text{ k}\Omega}{32 \text{ k}\Omega} \right) (15 \text{ V}) \\ &= (0.625)(15 \text{ V}) \\ &= 9.375 \text{ V} \end{aligned}$$

Therefore, the voltage, v_o across load resistor $20 \text{ k}\Omega$ is $\boxed{9.375 \text{ V}}$.